always double-check the specifications for your motor. It is also important to remember that while we are in essence using a form of PWM, we do not want to try to PWM the servo signal line using `analogWrite()` or we could damage our servo motor.

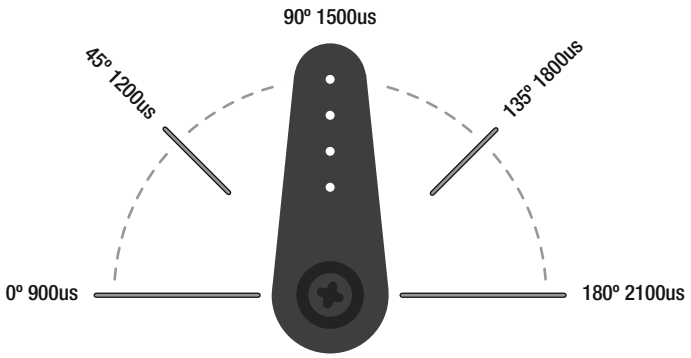


Figure 9-5. *Hobby servo rotation*

The Servo library can control up to a total of 12 servos on the Arduino Uno on any of the digital pins, although this functionality disables PWM on digital pins 9 and 10. Servos have three wires: signal, power, and ground. The yellow, white, or sometimes orange wire is for the logic signal and should be connected to one of the Arduino digital pins. The red wire is power and is connected to the +5v pin on the Arduino board. The black or brown wire is for ground and should be connected to the ground pin on the interface board. While we can control quite a few servos, they can consume current in excess of 100s of milliamps, so Arduino's power supply can't handle more than a few servos at a time. The supply voltage for servos should be in the +4.8v to +6v range, while the logic signal should remain within a +3–5v range. If using an external power supply, it is important that the ground wires share a common connection with the Arduino board. Figures 9-6 and 9-7 show how easy it is to connect a single servo to the Arduino board and we will then continue the discussion with some example code.

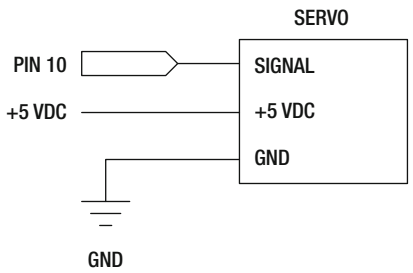


Figure 9-6. *Servo schematic*